

Comprehensive Survey On Intrusion Detection Techniques For MANET

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Abstract. Ad-hoc networks have been playing a vital role in various applications for many years. Dynamic Topology, Limited Resources, Open Environment, Lack of Infrastructure, Limited Bandwidth, Node Mobility are some of the key characteristics of the mobile adhoc networks and these unique characteristics post a significant threat to the security of the network. Encryption, authentication and firewall systems provide basic security features in Mobile Ad-hoc networks. However, the Intrusion Detection System (IDS) establishes extra security by preventing various attacks. Various IDS are built to provide better security features. In this work, the taxonomy of IDS based on different parameters are presented. This work compares performance of various IDS that are implemented using different techniques. At the end, this work concludes by indicating the research opportunities available for researchers to explore further.

Keywords: ad-hoc Network, intrusion detection system, manet, topology.

1 Introduction

- The wireless networks entirely changed the way of communication in the world. Wireless networks use air- medium to establish communication among nodes or devices. There are many wireless technologies available that include Bluetooth, Wi-Fi, 5G etc. [1, 2]. Applications of wireless technologies include surveillance and real-time multimedia applications, space and mission-critical applications and industry 4.0 applications. Wireless networks are divided into two parts, namely, Infrastructure-based Wireless Networks and Infrastructure-less Wireless Networks (IWN). The infrastructure-based wireless networks are dependent on a base station with a fixed number of nodes. They are expensive and predefined routes are used for data transmission. They cannot be used for hostile applications, such as earthquake prediction and military applications. Infrastructure-less networks do not have base stations and can be deployed without much infrastructure especially to mitigate natural calamities and

disasters. Differences between Infrastructure-based wireless network and infrastructure-less wireless networks are presented in Table 1.

Table 1. Comparison between Infrastructure-based and Infrastructure-less Wireless Network

Infrastructure-based wireless network	Infrastructure-less wireless network
Faster than Ad-hoc network.	Slower than Infrastructure networks.
More security options due to fixed base stations.	Security options are less due to the absence of a centralized station.
It is used to set a permanent network.	It is used to set up a temporary network.
Frequently used in hotel, airport lounges etc.	Frequently used in disaster zones, military, local networks, to establish communication among fixed groups of people etc.

Ad-hoc networks are classified into two classes based on mobility, namely, Mobile Ad-hoc networks (MANET) and Wireless Sensor Networks (WSN). In MANETs, each node acts as both router and host, movable and does not rely on fixed position. But the nodes in WSNs are immovable and stay where they are installed. Differences between MANETs and WSNs are listed in Table 2.

Table 2: Comparison between MANETs and WSNs

MANETs	WSNs
Dynamic Topology	Static Topology
Heterogeneous Network	Homogeneous Network
Power Rechargeable	Permanent Power
Unknown Destination	Known Destination
Peer-to-Peer to Communication	One-to-Many Communication

2 Manet Vulnerabilities

This section discusses the issues and challenges in the design and implementation of Infrastructure-less Wireless Networks (IWN). The factors that make IWN more vulnerable are discussed in this section. IWN has restricted physical protection and the nodes are either unprotected or loosely protected which makes the network vulnerable to security risk. The security of these networks containing unprotected nodes is more challenging and opens the door for more malicious nodes. Wireless links do not afford equivalent safety in transmitting the data when compared to wired links. There is no adequate line of protection which leads to unauthorized user access. Nodes are permitted to dynamically join or leave a network without any notification or intimation. Due to the absence of centralized management, it becomes very difficult to monitor the network traffic in the Ad-hoc networks mainly when it is on a large scale. This leads to high security risk [34]. Attackers

may introduce a Denial of Services attack (DoS) or Distributed Denial of Services attack (DDoS) that makes the nodes consume more power because of extra broadcast or high calculation. Various low-capacity links or nodes exist when compared to wireless networks which are open to external noise, interference, and other signals called bandwidth restriction [32].

Security attacks are classified into passive and active attacks [3]. In a passive attack, attackers gather information without affecting or altering the data. These types of attacks are difficult to identify. Active attacks are planned or predetermined against data Confidentiality, Integrity, and Availability i.e. CIA triad. Attackers encounter to change, drop, and block the packet transmitted over the network. The security attacks are further divided into two classes, namely, internal and external attacks [4], [34], [35]. Internal attacks are done in the same network on the compromised nodes and affect the normal routine of the network or system. External attacks are accomplished by external users. They belong to different network domains. A denial of Service (DoS) attack targets a network or server and floods it with network traffic. In a Black hole attack, the compromised or malicious node tricks by sending fake routing information, pretending that it has the shortest path to the destination, and drops all the packets instead of forwarding them [34]. In a Wormhole attack, the attacker receives the packet and tunnels the received packet to another point in the same network due to this the routing can be disturbed [34]. Man-in-the-middle attack is an attack where the attacker sits in between two nodes and sniffs the information shared by two nodes [35]. There are two security mechanisms for mobile ad-hoc networks.

1. Security Prevention Technique: Primary defense mechanism against external attack is by cryptographic techniques. It contains integrity and data authenticity.
2. Security Detection Technique: The secondary defense mechanism was initiated when the primary defense failed. Intrusion Detection System (IDS) protects against any security threats, risks, and vulnerabilities (CIA triad). IDS desires to search for any abnormal or unwanted activities in a network and an Intrusion Prevention System (IPS) prevents them before they harm the network.

An Intrusion Detection System (IDS) is either a hardware or software-based system to search or trap any kind of intrusion at its previous stages. It combines tools, methods, and resources that identify the intrusions. IDS is built around a single device or around a complete network. IDS can be used both in wired and wireless networks. There are different methodologies used to develop schemes for IDS in ad-hoc networks based on statistical, heuristic, state, rule, and reputation. Several techniques have been proposed to investigate intrusion detection in Ad-hoc networks [32].

3 Manet Vulnerabilities

The security of an ad-hoc network is mainly based on various types of attack types, security challenges, and requirements. The classification of ad-hoc network security is given in Figure 1.

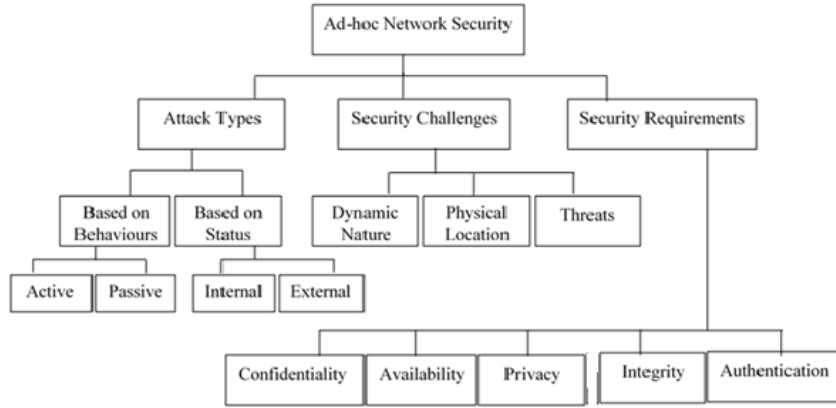


Fig. 1. Taxonomy of Ad-hoc Network Security

2.2 Security Requirements

Security needs should be considered, checked, and implemented correctly to provide better security to the ad-hoc network. MANET requirements differ with various environments and applications and the necessary goals are discussed in this section [34, 35]. In the confidentiality sector, every node must access a specific service in the network. It ensures that the sensitive data are accessed by authorized users and also protects sensitive data from unauthorized users. Information leakage in mainly financial and military transactions has dangerous and irrecoverable damages. Confidentiality can be improved with authentication. Authorization allocates different access rights to various types of users by having a multi-authentication factor. In the Availability sector, every node must have access to all data and services in the network. It assures that the intended services are available to use. The attack domain comprises adversaries that promote traffic by interfering the communication with various attacks such as DoS or DDoS attacks on routing protocol, physical channels, network disconnection, and key management issues. Challenges in the availability sector arise due to MANETs' dynamic topology and open boundary. Information that is particular to a specific system, network, or organization is called private information, e.g. system-id, password, location, and biometrics. Maintaining access to private information that no one can access without proper permission is called Privacy. In the Integrity

sector, only authorized nodes can edit or make changes to the information. It confirms that nobody could intrude and damage the data transferred via an Ad-hoc network. The receiver of the data must confirm that the sender of the data is authenticated. A compromised system may be likely to send damaged data. Authentication confirms that the communication between two entities is authorized. Communication without any authentication leads to unauthorized access and also opens the door for malicious attacks.

3.2 Security Challenges

Different features of ad-hoc networks pose a few challenges that are related to security and it might be considered while developing security protocols for these networks. Various attacks extend from passive to active because of the open nature of wireless options. Attacks on these networks get affected from all sides whereas wired networks are guarded by gateways and firewalls. Threats may lead to damages which include node impersonation, data leaks, and message alterations. Some of the threat information is provided in Table 3.

Table 3: List of various threats

Year	Threats	Description
Early 2000's	Hackers	Information is accessed by unauthorized users
2003	Intrusion Worms	Malicious software combining intrusion tactics and worm-like propagation, rapidly spreading across networks to exploit vulnerabilities and compromise systems
2005	Viruses, Spyware	Virus-Malicious software that infects the systems and self replicate Spyware-Gather sensitive information of users
2007	Botnets Spam	Steal data by spamming, DDoS attack or credential theft
2008	Malicious URLs	Phishing of messages which leads to security breaches
2011	Malicious Apps	Apps automatically connects to any services without our knowledge
Today	Advanced Persistent Threats	Network attack done by hackers who get network gains.

Network security solutions with fixed configurations are not sufficient for the dynamic network topology. Decentralized security systems are advisable to prevent problems. When it comes to physical attacks, all nodes are independent in ad-hoc networks and also have a low level of physical security due to dynamically changing topology and lack of centralized monitoring. So, nodes in Ad-hoc networks can be compromised easily. In an ad-hoc network, a malicious node can

cause several attacks which are very difficult to identify since tracking such nodes in a large number of nodes in an ad-hoc network is a difficult process

3.3 Types of Security Attacks

Users are exposed to attacks due to dependency on the global network. Attacks on mobile ad-hoc networks can be categorized based on the attacker's status and behavior [7].

1. User-based Status: Attacks on ad-hoc networks can be divided into internal and external attacks. The attacks which are happening in the same network are called internal attacks. It can be performed by either users or nodes in the same network to access resources. On the other hand, external attacks are performed outside the network by unauthorized nodes.
2. User-based Behaviour: Attacks are classified into passive and active based on behavior. In Passive attacks, attackers gather information or monitor and eavesdrop without affecting or altering the data even indirectly. Active attacks are planned or predetermined against data confidentiality, integrity, and availability (CIA triad). This kind of attack disrupts the resources and changes the routing paths between source and destination. The damages include modification of network behavior and false data incorporation.

4 Architecture Of Intrusion Detection System (IDS)

Threats may infect systems or networks by breaking the primary line of protection i.e. privacy, routing, authentication, access control, key generation, and leaving the system or network unsafe [32]. MANETs are not able to identify any intrusions or attacks inside its network. The second line of protection includes an Intrusion Detection System (IDS) which detects attacks with greater assurance and an Intrusion Prevention System (IPS) which stops them before causing big damage. IDS is a complete network that performs the following activities: i) scans tasks in a single system or group of systems, ii) makes analysis of the inbound network traffic, iii) clashes incidents are filtered out which are against the system security policy, and iv) concludes the activity as an unauthorized activity from a malicious entity. The objective of IDS is to automate the intrusion detection in the network that tries to interrupt the Confidentiality, Integrity, and Availability of the network i.e. CIA triad by any kind of attack. IDS observes the behavior of networks, identifies the misbehaved activities, and initiates the alarm when malicious activity is identified.

Existing IDS cannot provide a good solution to all types of attacks and every system is constructed to provide a good security solution. Essential modules of IDS are data collection, data pre-processing, intrusion recognition, and report generation as shown in Figure 2. The traffic data is collected from a single machine or network which consists of both normal and abnormal data. The collected data has unwanted and unordered data which should be cleaned and modified. The Intrusion Detection module is responsible for identifying malicious activities based on the rules provided in the knowledge base. IDS enhanced edition is provided by the administrator and contains a profile of already known intrusions and compares with the newly arriving data against the past intrusion profiles. In this way, any abnormal behavior from audit data can be identified. The report module generates the alarm when intrusion is detected. The nature of the attack is sent to the administrator and stored in the database for future reference. The performance measure of IDS is a difficult task due to many factors. However, for each detected activity the degree of IDS accuracy can be measured and explained as shown in Table 3.

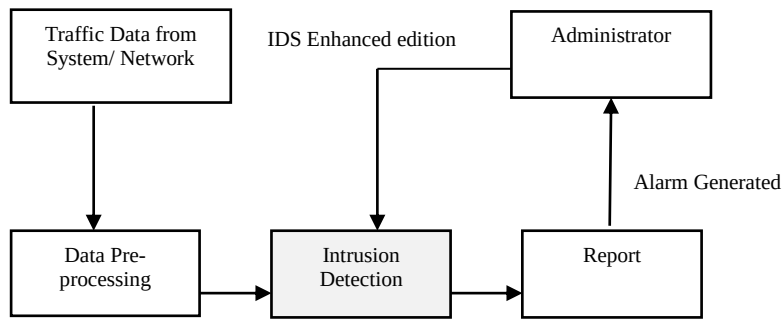


Fig.2. Intrusion Detection System

True Positive (TP): The identification is said to be truly positive when intrusion is successfully identified. **True Negative (TN):** The identification is said to be true negative when it predicts the negative result correctly. **False Positive (FP):** In this state, the IDS detects it as malicious but the activity is legal. **False Negative (FN):** In this state, IDS shows the activity is legal but the activity is malicious. False Negative impacts MANET severely. History of IDS started three decades ago with the trend of network usage by users. The term Intrusion Detection was introduced by James P. Anderson in the U.S Air force during the 1980s. He had introduced automated IDS reports with help of Computer Security hazard Surveillance and Monitoring. The first application built method is used as an anti-virus. It has scanned a list of known threats and compared them with network traffic. During the late 1980's, system experts all over the world had adapted intrusion detection systems. During 1990's, IDS technology improved significantly and detection rate increased. Advantage of IDS is, to identify unknown attacks that are not available in the signature list and able to alert system administrators. Details of IDS in various aspects are presented in Table 5. It describes the evolution of intrusion detection systems.

Table 4: Evolution and Timeline of Intrusion Detection System

Timeline of the IDS	Timeline	Architecture	Type of Attacks	Remarks
IDSX [36] , 2007	2007	Extended Hierarchical Architecture	Routing misbehavior	Expensive in terms of resources and energy utilization.
Neural Network & Watermarking Technique [12], 2007	2007	Self-Organizing Map	Routing behavior attacks	Additional training overhead leads to energy waste to find these attacks.
CONFIDANT [37]	2002	Distributed and Cooperative	Packet drop attacks	Network damage due to wrong assumptions by network nodes.
HIDS [32]	2008	Distributed and Collaborative	Black Hole Attack and Packet drop attacks	Test is needed before secure routing construction.
BHIDS- A black hole IDS [41]	2010	Hybrid	Black hole attack	Able to identify only black hole attack.
Leader Election Model [38]	2008	VCG model	Resource utilizing attack-selfish nodes	Selfish nodes do not drain energy to run IDS service, the percentage of packet analysis reduces with the help of time.
SCAN [39]	2002	Distributed and Collaborative	Routing misbehavior and packet forwarding misbehavior	Network services are temporarily halted during detection.
OCEAN	2003	Standalone IDS	Rushing attack	It is not effective to find misleading nodes.
A System of Umpires [40]	2010	Collaborative IDS for double and triple users	Packet dropping attacks	Umpires are not permanent. Election is needed to re-elect the leader. Energy consumption is high.
Feed-forward neural network, Support vector machine. [42]	2016	Distributed and Collaborative	Grey Hole and Rushing Attacks	Fuzzy Petri Nets can be added to enhance the performance
Adaptive Neuro Fuzzy Inference System, Particle Swarm Optimization	2020	Hybrid	Black hole attack	Able to identify only black hole attack.

[43]	Proportional Overlapping Scores (POS) method, Artificial Neural network	2022	Collaborative	Rushed Black and Gray Hole attacks	Node mobility is not considered.
[44]					

Some important issues encountered by IDS in ad-hoc networks are discussed and these challenges must be resolved. The frequent changes in topology offer IDS with flexible architectures. It is desirable to develop co-operative and distributive IDS due to a lack of central management. Ad-hoc networks are vulnerable since they lack proper boundaries and allow outside attacks. The bandwidth of Ad-hoc networks is limited. This forces the IDS to minimize the interaction between the nodes as much as possible. IDS has to follow cryptography since the information should not be revealed easily when agents are compromised. The power source is limited in ad-hoc networks. So, IDS uses a low computational approach. However, the initial IDS has a few problems. *i)* It could send alert messages only for known attacks; *ii)* resources are drained frequently due to continuous scanning and updates of the signature list. Some of the IDS algorithms that are created for MANETs are listed in Table 5.

5 Classification of Intrusion Detection Techniques

Many methodologies are used to detect intrusions in mobile ad-hoc networks. This section deals with the most important and well-known techniques used to find the intrusions. These techniques include signature-based, anomaly based, and hybrid based. Figure 3 shows the classification of intrusion detection systems.

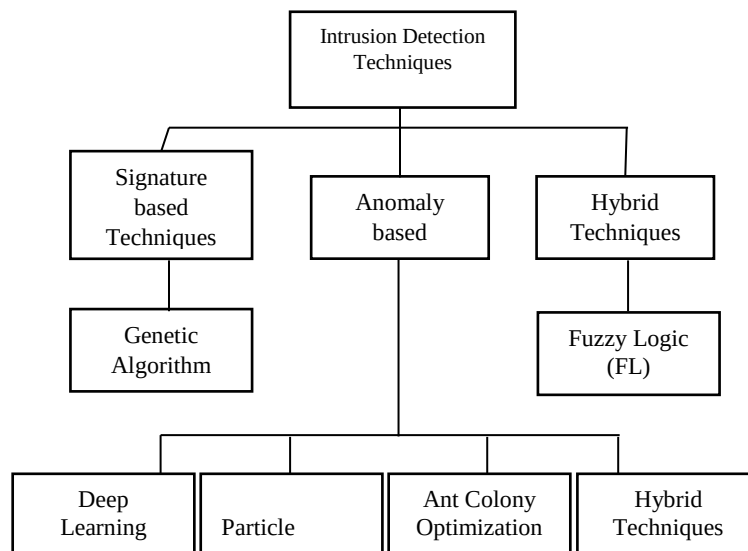


Fig. 3 Classification of Intrusion Detection System

5.1 Signature based IDS

A signature is a pattern of a unique network traffic change. All the patterns are stored in the database and the new patterns are reviewed with the previous records. Signature-based technique is also called a knowledge-based technique as it uses previously recorded patterns to check the intrusions. It works like an antivirus which is based on the recorded virus patterns but fails in detecting new threats.

5.1.1 Genetic Algorithm based Intrusion Detection

Natural selection is the base for genetic algorithms. It simulates the survival of the fittest or natural selection process. A genetic algorithm selects individuals from a random population and produces the children for the next generation. Shingo et. al have proposed a network intrusion detection model based on fuzzy class association rule mining using Genetic Network Programming (GNP) [9]. This method finds the differences between the behavior of the unauthorized connection and the normal connection using a fitness function. This technique is progressively optimized and uses direct graph structures. It contains databases that include both discrete and continuous attributes. Association rule is generated based on the nature of the attack to identify threats. The methodology proposed in [9] captures both misuse and anomaly types of attacks. Genetic algorithm and Support Vector Machine (SVM) are the building blocks of the proposed IDS. It uses feature selection, weight, and parameter optimization for effective classification by SVM. Crossover and mutation probability of GA is applied for optimization. Fitness value and population evolution algebra are used for effective feature selection that reduces error rate, increases true positives, and reduces classification time subsequently. Ramy et. al [11] have proposed a framework for intrusion detection using hybrid nested genetic-fuzzy algorithms. The Hybrid Nested Genetic-Fuzzy Algorithm framework is used to produce outputs that have normal and abnormal data. A genetic Convolutional neural network for intrusion detection systems is used for network-based IDS [12]. It can be achieved using genetic algorithms for subset feature selection and fuzzy C-means clustering [9].

5.2 Anomaly based IDS

This technique is also called behavior-based IDS. The normal behavior is recorded and derived over time by monitoring various activities such as hosts, network connection and regular activities. The specific threshold is marked and any deviation from this value is called an anomaly. An alarm is generated when there is a large amount of deviation. It is preferred over signature-based models since the new intrusions are detected effectively. A profile is created from attributes such as the usage of the processor, the number of emails sent and the number of login attempts failed. The model has two phases: training and detection. In training, the behavior is learnt. When any intrusion occurs, the IDS checks with the usual behavior. If a particular behavior does not match with a normal profile, it alarms and marks the behavior as abnormal. The unknown behaviors are alarming and they are sent to the expert team and stored in the database for future reference.

5.2.1 Deep Learning based Intrusion Detection

Sydney et. al have proposed an intrusion detection system that uses feature extraction with deep learning [13]. There are two types of feature extraction, namely, wrapper-based and filter-based. Deep Learning (DL) is used in wrapper-based feature extraction. It uses DL models to classify and measure the performance of those features. Feed-forward deep neural networks are used to produce a reduced optimal feature vector. This proposed model in [13] is applied for both binary and multiclass classification. Mohammad et. al has proposed an intrusion detection DL model for a big data environment using a hybrid deep learning approach [15]. Models such as Convolutional Neural Networks (CNN), Deep Brief Networks (DBN), and Weight-Dropped Long Short-Term Memory (WD LSTM) networks are used to detect intrusions. CNN extracts and gathers information about the meaningful features from the data traffic present in a network that exploits the speed. LSTM helps to retain the dependencies that are long-term and prevents the gradient vanishing problems. DL models are evaluated with the help of intrusion datasets such as UNSW-NB15 and ISCX2012 [17]. Wisam et. al [16] have proposed network intrusion detection for deep learning architecture using a double PSO metaheuristic. More deep learning models are pre-trained by a double PSO-based metaheuristic algorithm to achieve high performance. This revised algorithm consists of two different levels upper and lower levels. The upper level is used to select optimal features from a given subset and later hyper-parameter is used to increase detection accuracy from a reduced data set. The proposed model in [16] used the Friedman statistical test and two different ranking methods to conduct the test. The detection accuracy rate increased from 4% to 5% and the false alarm decreased. Arwa et. al have proposed a detailed survey on anomaly-based IDS using deep learning approaches [17]. The proposed methodology in [17] provides a taxonomy of deep learning for IDS with different aspects. Zhendong et. al [18] proposed an IDS model using Semantic re-encoding [18]. It increases the semantics between the traffic so that the abnormal traffic can be easily distinguished and identified. The robustness and performance of the models are improved by using the proposed methodology in [18] and it identifies web injection attacks effectively. A survey and taxonomy of deep learning models are presented by Sunanda et. al in [19]. Models such as feed-forwarding neural networks, LSTM networks, autoencoder, and DBM are used for evaluation. Four datasets i.e. Two legacy KDD 99, and NSL-KDD and two modern CIC-IDS2017 and CIC-IDS2018 are used for training. Mohamed et. al has proposed IDS by implementing DL models that are trained by 35 well-known cyber datasets which are categorized into seven categories, namely, Network traffic-based, Internet traffic-based, Virtual Private Network-based, Electrical network-based, Android apps-based, IOT traffic-based and Internet-connected devices-based datasets [20]. These seven deep learning approaches come under two models such as deep discriminative models and generative unsupervised models. The robustness and accuracy of IDS is improved by using these different DL models.

5.2.2 Particle Swarm Optimization based Intrusion Detections

Particle Swarm Optimization (PSO) is a method that effectively optimizes a problem just by trying to improve the solution of a candidate with respect to a given measure in an iterative way. Botnet threat is a huge risk for internet users. The BD-PSO-V model is proposed in [21] to detect botnets using a hybrid technique. This technique is a combination of particle swarm optimization and a voting system. PSO is used to select appropriate features and a voting system is used for the classification of samples. Datasets such as ISOT [10] and Bot-Iot [11] are given as inputs for PSO and Deep Neural networks, and SVMlib, and Decision tree are given as inputs for the voting system. Decision-making is done based on maximum votes. Hadeel Alazzam et al. [22] have proposed wrapper method-based feature selection to identify intrusion detection using an inspired Optimizer (PIO). A robust IDS is developed with a reduced feature set. Discrete optimization process i.e. cosine similarity is applied to swarm intelligent algorithms to solve problems. Yu Xue et al. [23] have proposed solutions for large-scale feature problems that include feature selection. It is achieved by using multiple classifiers. Whale optimization approach, Grasshopper optimization approach with evolutionary population dynamics, and Binary dragonfly algorithm are some of the algorithms used for feature selection. Four classifiers are used to evaluate this model. Bach Hoai Nguyen et al. [24] have proposed a new survey on feature selection in classification problems using swarm intelligence. Swarm intelligence adopts an easy mechanism to filter relevant data from a large data set. This survey considers three SI algorithms, namely, ABC, PSO, and ACO. Ankit Thakkar et al. [25] have proposed a survey on swarm and evolutionary algorithms for IDS. Swarm intelligent algorithms are taken into consider for calculating parameter optimizing, feature optimizing, and attack classification. Mohammad Mehedi Hassan et al. [15] have proposed IDS in a particular type of network called wireless mesh networks which use feature selection methods. Those methods use Whale optimization algorithms and genetic operators. Modification of WOA solves the Convergence problem. To increase the search process, modification of WOA happens along with crossover operation. ADFA-LD and CICIDS2017.

5.2.3 Ant Colony Optimization based Intrusion Detections

The Ant Colony Optimization algorithm is a technique used to solve computational problems. This algorithm is used to reduce the computations by using graphs. Artificial ants are made as agents mimicking the behavior of original ants. Ahamed Fayeez Tuani et al. [26] have proposed solutions for the traveling salesman problem using heterogeneous adaptive ant colony optimization (ACO) with a 3-opt local search. Good parameters are chosen by adaptive tuning methods that include ant colony optimization to reach near-optimal solutions. Fitness can be improved by a 3-opt local search heuristic strategy. Yassir Aberni et al. [27] have proposed a palm vein recognition method for personal authentication and identification. This is based on the scheme of competitive coding which uses a Multiscale Local Binary Pattern (MLBP) along with Ant Colony Optimization (ACO).

5.2.4 Reinforcement Learning based Intrusion Detection

Reinforcement Learning based Intrusion detection system helps in identifying the network layer attacks happening in uncertain environments. Double layer attack detection framework [45] based on Deep Q- learning and Deep Autoencoder to detect False data injection attack. The algorithm produces 98.20% detection rate and False Positive rate of 2.9%. Reinforcement based malicious node [46] is devised to mitigate network layer attacks in adhoc networks. The proposed method outperforms the state of art techniques in terms of Accuracy and False positive Rate. Reputation based opportunistic routing method using Q-learning [47] is developed to mitigate the network layer attacks such as black hole and grey hole attack. This method is efficient in terms of packet loss, average end to end delay and energy level. Q-learning based technique is proposed for secured routing [48] in On demand Distance Vector protocol which enables the NS-3 simulator to consider more number of neighboring nodes for simulation. This technique makes the network more reliable by detecting the black hole attack with high accuracy.

5.3 Hybrid Intrusion Detection System

Fuzzy logic is a mathematical theory that allows the processing of multiple truth values using a single variable. It also exists in the real biosphere. It is implemented in Intrusion Detection Systems as it has the ability to merge the data of different sources frequently. Selvakumar K et al. [29] have proposed IDS which is based on the Fuzzy rough set-based nearest neighbor classification. This model is used to train the classifier. The number of features is kept small to balance between optimal detection rate and performance in the real world. Allen's intervals of algebra operators removed the redundant features. False positives and false alarm rates are reduced in this model. Intrusion detection is developed for real-time using neuro-fuzzy classifiers and implemented on GPUs [30] A high detection rate is achieved by an Enhanced fuzzy min-max neural network. Minimum training and recall time are enough. Manimurugan et al. [31] have proposed an IDS that has been based on the crow search optimization algorithm which is in combination with an Adaptive neuro-fuzzy inference system. Here the fuzzy rules are not imposed in the input information and the hybrid learning utilizes the data pairs of input and output. Comparison among intrusion detection mechanisms are tabulated in Table 4 and an overview of the IDS classification mechanism is presented in Table 5 and a list of datasets that are used for IDS is listed in Table 6.

Table 5: Advantages and Disadvantages of IDS Mechanisms

Detection Mechanism	Advantages	Disadvantages
Signature Based	• This simple method is more effective and accurate in capturing known intrusions. • Detection is fast. • Fast and frequent protection.	• Unknown attacks are unable to be detected. • Lack of understanding in states and protocol. • Frequent updates are unpractical. • More time is needed in maintaining the knowledge.
Anomaly Based	• Unknown attacks are handled here. • Not much dependent on System software such as Operating systems. • Provide facility for detection of privileged abuse.	• Because of frequent change in observed events, profile maintenance may be poor. • While rebuilding the profile with changes in the normal behavior, the service is stopped. • Timely provision of alarm is poor
Hybrid	• According to the requirements, the IDS is developed. • Any type of mechanism can be adopted due to its hybrid nature.	• Not suitable for devices that are more concerned with memory usage. • Powerful computations lead to high consumption of power.

Table 6: Overview of IDS Classification Mechanisms

Primary IDS Techniques Used	Additional Schemes used	Parameters Considered
Genetic Algorithm [9]	GNP-based Fuzzy Class Association Rule	Detection Accuracy, Low False Alarm Rate
	HNGFA Framework, OGFA & IGFA	NIDS Detection Accuracy
	IFS, CNN Extractor & BG Classifier and 5-fold CV	BG Classifier Over fitting Problem & Performance
Deep Learning [13]	Feed Forward Deep Neural Networks, Extra Trees (ET) algorithm, WFEU-FFDNN, CNN-WDLSTM IDS Large-scale big data	Binary and Multi-class classification, NIDS and real-time IDS
	SRDLM method for re-encodes, SRDLC	Accuracy and Robustness of algorithm, Web character injection network attack
Particle Swarm Optimization [21]	WOA and genetic operators Performance	Binary and Multi-class classification
	Friedman statistical test & Ranking methods	BD-PSO-V Botnet Threat
Ant Colony Optimization [26]	3-opt local search	Multi-scale local binary pattern and Low and small budget function
Fuzzy Logic [29]	Allen's intervals algebra operators	Redundant data removal
Reinforcement Learning	Deep Q-Learning with Deep Autoencoder	Detection Rate, False Positive Rate
	Reinforcement learning-based malicious	Accuracy and False

nodes	Detection Rate
Reputation Opportunistic Routing based on Q-learning (RORQ)	Packet loss, Average end-to-end delay, Energy efficiency
Q-learning in Ad-hoc On-demand Distance Vector Protocol (AODV)	Accuracy and Packet Drop Ratio

Table 7: Important Dataset for IDS

KDD CUP 99, NSL- KDD, UNSW-NB15, AWID, CICIDS2017, CIC-IDS2018, ADFA-LD, CSE-CIC-IDS2018, ISOT, Bot-IoT and UCI ML Network Traffic-Based , Electrical Network-Based, Internet Traffic-Based, Virtual Private Network-Based, Android Apps- Based, Iot Traffic-Based And Internet-Connected Devices-Based [20]
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6 Conclusions

The paper presents a survey on the features of Mobile Ad-hoc Networks and their challenges. Various types of attacks, such as active attacks, passive attacks, wormhole, and black hole attacks, occur in different scenarios within ad-hoc networks. The Intrusion Detection System (IDS) is elaborated, including its working modules, challenges faced, requirements, and its role in ad-hoc network security. Detection techniques, datasets used, associated schemes, tools, merits, and demerits which are supported by figures and tables are present in this survey. The interactive architecture and key information of IDS types have been explained based on the detection mechanisms provided by the taxonomy of IDS. Classification of IDS includes Signature-based, anomaly-based, and Hybrid techniques. IDS detection methodologies containing various schemes are summarized. Many issues in previous works related to IDS are pointed out and elaborated. Strong points and limitations are clearly stated for the work of each reference. Clear comparisons are given through detailed tables.

In the future, it would be a great idea to increase the range of different attacks in a single attempt and consider privacy issues when installing the system in a mobile ad-hoc network. Dealing with a multi-level or hybrid approach poses a high computational cost for attaining results. Future research aims to adopt mechanisms that achieve cost-effectiveness. Shortening the expected time for feature training and testing is crucial. Reducing false alarms plays a significant role in IDS since these issues degrade overall performance. Existing intrusion detection systems lack sufficient accuracy in real-time threat detection. Thus, new research focuses on real-time IDS to identify attacks promptly.

7 Acknowledgement

The research paper was completed with the help of various individuals to whom we express our sincere gratitude. From our hearts, we appreciate Mrs. R. Manimegalai, head of the Department of Computer science and Engineering at PSG Institute of Technology and Applied Research, Coimbatore, India, for their guidance throughout the research process. Finally, we would like to acknowledge the financial support provided by PSG Institute of Technology and Applied Research, Coimbatore, India.

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